

Liuyin Yang

PhD Researcher · EEG Foundation Models & Brain–Computer Interfaces · KU Leuven

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I build **EEG foundation models** for brain–computer interfaces: transferring large models pretrained on scalp EEG to **intracranial recordings** (CORTEG, NeurIPS 2026), testing when foundation models beat traditional decoders (ICLR 2026), and decoding auditory and motor signals in real-time BCI systems. Winner of the **NeurIPS 2025 EEG Foundation Challenge**.

Education

2021 – now **PhD, Biomedical Sciences (Medicine)** · KU Leuven, Belgium
EEG-based brain–computer interfaces. Advisor: Prof. Marc M. Van Hulle.

2020 – 2021 **Advanced Master of Artificial Intelligence** · KU Leuven
Big Data Analytics · Cum laude (75.33%)

2019 – 2020 **MSc, Electronics & ICT Engineering Technology** · KU Leuven
Cum laude (73.85%)

2017 – 2019 **BSc, Electronics Engineering** · KU Leuven
Cum laude (72.58%)

2015 – 2017 **BEng, Electrical Engineering & Automation** · Chongqing University, China
GPA 3.3 / 4

Research Experience

2021 – now **PhD Researcher** · Computational Neuroscience Group, KU Leuven

- Cross-modality transfer of scalp-EEG foundation models to intracranial (ECoG) recordings with an electrode-aware spatial adapter and dual-stream tokenizer (CORTEG, NeurIPS 2026).
- Pretrained a large spatiotemporal transformer on EEG with masked autoencoding; scaled pretraining on the VSC Tier-2 HPC; benchmarked EEG foundation models against traditional decoders across BCI tasks (ICLR 2026).
- 1st place, NeurIPS 2025 EEG Foundation Challenge (cross-task and cross-subject EEG decoding).
- WaveNet-based auditory EEG regression model, 2nd place in the ICASSP 2023 Auditory EEG Grand Challenge.
- Online motor-imagery BCI for real-time navigation in Google Street View; binocularly incongruent SSVEP BCI in VR (Meta Quest Pro).

2019 – 2020 **Master's Thesis Researcher** · KU Leuven, Engineering Technology
Real-time adaptive-filter speech enhancement (noise reduction) on a DSP board, deployed in an IoT communication system.

Awards & Honors

2025 **1st Place — EEG Foundation Challenge 2025** · NeurIPS 2025 Competition Track (From Cross-Task to Cross-Subject EEG Decoding)

2023 **2nd Place — ICASSP 2023 Grand Challenge** · Auditory EEG Decoding Challenge

2019 – 2021 **Cum laude in three consecutive KU Leuven degrees** · Advanced Master AI, MSc Electronics & ICT, BSc Electronics

Publications

CONFERENCE PAPERS

NeurIPS 2026 **CORTEG: Foundation Models Enable Cross-Modality Representation Transfer from Scalp to Intracranial Brain Recordings.** L Yang*, Q Sun*, B Van Dyck, EC Merino, MM Van Hulle (*equal contribution). *Conference on Neural Information Processing Systems (NeurIPS) 2026,* poster.

ICLR 2026 **Are EEG Foundation Models Worth It? Comparative Evaluation with Traditional Decoders in Diverse BCI Tasks.** L Yang, Q Sun, A Li, MM Van Hulle. *International Conference on Learning Representations (ICLR) 2026,* poster.

ICORR 2025 **On the Impact of Proprioception in EEG Representations and Decoding during Human-Hand Exoskeleton Interaction.** Q Sun, EC Merino, **L Yang**, A Faes, MM Van Hulle. *International Conference on Rehabilitation Robotics (ICORR) 2025.*

ICASSP 2023 **Decoding Auditory EEG Responses Using an Adapted WaveNet.** B Van Dyck, **L Yang**, MM Van Hulle. *IEEE ICASSP 2023, Auditory EEG Grand Challenge (**2nd place**).*

JOURNAL ARTICLES

2024 **Unraveling EEG Correlates of Unimanual Finger Movements: Insights from Non-Repetitive Flexion and Extension Tasks.** Q Sun, EC Merino, **L Yang**, MM Van Hulle. *Journal of NeuroEngineering and Rehabilitation, 2024.*

2024 **Binocularly Incongruent, Multifrequency-Coded SSVEP in VR: Feasibility and Characteristics.** **L Yang**, Q Sun, MM Van Hulle. *Journal of Neural Engineering, 2024.*

2024 **Sea-Wave: Speech Envelope Reconstruction from Auditory EEG with an Adapted WaveNet.** **L Yang**, B Van Dyck, MM Van Hulle. *IEEE Open Journal of Signal Processing, 2024.*

2023 **Real-Time Navigation in Google Street View Using a Motor Imagery-Based BCI.** **L Yang**, MM Van Hulle. *Sensors, 2023.*

2021 **Chronic Study on Brainwave Authentication in a Real-Life Setting: An LSTM-Based Bagging Approach.** **L Yang**, A Libert, MM Van Hulle. *Biosensors, 2021.*

WORKSHOP PAPERS

ICLR 2026 W **Do We Need Domain-Specific Time-Series Models? Insights from EEG Classification Benchmarks.** **L Yang**, Q Sun, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026,* **poster**.

ICLR 2026 W **Learning Robust EEG Representations with a Large Spatiotemporal Transformer as a Foundation Model.** **L Yang**, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026.*

ICLR 2026 W **Large EEG Foundation Model Learns Informative Low-Frequency Representations from Intracranial Brain Signals.** Q Sun, **L Yang**, EC Merino, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026,* **poster**.

PREPRINTS & DATASETS

2026 **Characterization of Speech Imagery in Scalp EEG and Comparison with Motor Imagery.** B Van Dyck, **L Yang**, Q Sun, A Li, MM Van Hulle. *arXiv preprint* `arXiv:2606.12223`.

2025 **CoMET: A Contrastive-Masked Brain Foundation Model for Universal EEG Representation.** A Li, Z Wang, **L Yang**, Z Wang, T Xu, H Hu, MM Van Hulle. *arXiv preprint* `arXiv:2509.00314`.

2026 **A Multi-Paradigm EEG Dataset for Multi-Task BCI Applications.** **L Yang**, B Van Dyck, Q Sun, A Li, MM Van Hulle. *KU Leuven Research Data Repository,* `doi:10.48804/3EUSTS`.

Talks & Teaching

Dec 2025 **Spotlight talk — EEG Foundation Challenge 2025** · NeurIPS 2025 Competition Track, San Diego

2025 **EEG Foundation Models** · IEEE NER 2025 Workshop

2025 **Scaling EEG Foundation Model Pretraining** · KU Leuven / UHasselt Tier-2 HPC User Day

recurring **Teaching Assistant** · Brain–Computer Interfaces (MSc course), KU Leuven

Skills

Deep learning	PyTorch, distributed training (DDP), transformers, self-supervised learning
EEG & lab	EEG experiment design and data collection, signal processing, Unity (VR — Meta Quest Pro), Android Studio
Programming	Python, MATLAB, Octave, Java, C / C++ / C#, Assembler
Hardware	Vivado (FPGA), DSP board (C5515 eZdsp), Arduino, Eagle (PCB), microcontrollers
Languages	English — proficient · Chinese — proficient · French — intermediate